House Renting System

Technology Stack (Architecture & Stack)

Project Design Phase-II

Project Information

- Date: 8 March 2026
- Project Name: House Renting System
- Frontend: React.js
- Backend: Node.js (Express)
- Database: MongoDB / MongoDB Atlas

Technical Architecture

- The system follows a modern web architecture:

- • Frontend built using React.js
- • Backend API developed using Node.js with Express
- • MongoDB used as the database
- • Users can search houses and owners can list rental properties
- • REST APIs connect frontend and backend

Components & Technologies

S.No	Component	Description	Technology
1	User Interface	User interacts with system	React.js, HTML, CSS, JS
2	Backend API	Handles requests and authentication	Node.js, Express.js
3	Business Logic	Property listing & booking logic	Node.js Services
4	Database	Stores users and property data	MongoDB
5	Cloud Database	Managed database service	MongoDB Atlas
6	File Storage	Store property images	Cloud Storage / Local

Application Characteristics

S.No	Characteristic	Description	Technology
1	Open Source Frameworks	Frameworks used in development	React.js, Node.js, Express
2	Security	Authentication and encryption	JWT, bcrypt, HTTPS
3	Scalable Architecture	Designed to scale with users	REST APIs, modular backend
4	Availability	Cloud hosting with high uptime	AWS / Cloud hosting
5	Performance	Optimized queries and APIs	MongoDB indexing